02-EDA Exploratory Data Analysis - Pitt Dataset

We have new datasets from DementiaBank. Now we explore the English - Pitt dataset

In [1]:
```
# import required packages
import pandas as pd
import os
from wordcloud import WordCloud, STOPWORDS, ImageColorGenerator
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
import re
```

In [2]:
```
# labels
# AD = Alzheimer's Disease (positive class)
# CN = Cognitively Normal (negative class)
```

In [3]:
```
path_ad = "../data/TalkBank downloads/Pitt/Dementia/Cookie/"
path_cn = "../data/TalkBank downloads/Pitt/Control/Cookie/"
```

In [4]:
```
train_data = {"ad":[],
              "cn":[]
             }

file_names = {"ad":[],
              "cn":[]
             }
```

In [5]:
```
def line_clean(line):
    """
        Clean line from extra information like "*PAR: ", "\x1530_5640\x15", etc
    """
    result = line
    if result.startswith("*PAR"):
        result = result.replace("*PAR:\t", "")
        result = re.sub("?.*?", "", result)
        result = re.sub("\[.*\]", "", result)
        result = result.strip("&-+<").strip()
    else:
        result = ""
    return result

s = '*PAR:\tthe scene is <in the> [/] in the kitchen . \x1530_5640\x15\n'
assert line_clean(s) == "the scene is <in the>  in the kitchen ."
```

In [6]:
```
def read_data(path):
    """
        get data from "txt" files from a specified folder (ad/cn)
        returns list of text (strings)
    """
    result = []
    filenames = []
```

```
    # get only txt files
    files = [item
             for item in os.listdir(path)
             if item.endswith("cha")
            ]

    for fname in files:
        print("Processing files =", fname) #, end="\r")
        with open(path + fname, encoding="utf-8") as f:
            lines = f.readlines()
            print("Line counts =", len(lines)) #, end="\r")
            # apply filter to clean text
            lines = [line_clean(line)
                     for line in lines
                     if line.startswith("*PAR")
                    ]
            print("After cleaning, Line counts =", len(lines)) #, end="\r")

            text = "\n".join(lines)
            result.append(text)
            filenames.append(fname.replace(".cha", "")) # remove file extensions

    return (result, filenames)
```

In [7]:
```
train_data["ad"], file_names["ad"] = read_data(path_ad)
train_data["cn"], file_names["cn"] = read_data(path_cn)
```

```
Processing files = 001-0.cha
Line counts = 85
After cleaning, Line counts = 10
Processing files = 001-2.cha
Line counts = 125
After cleaning, Line counts = 13
Processing files = 003-0.cha
Line counts = 172
After cleaning, Line counts = 23
Processing files = 005-0.cha
Line counts = 73
After cleaning, Line counts = 9
Processing files = 005-2.cha
Line counts = 92
After cleaning, Line counts = 7
Processing files = 007-1.cha
Line counts = 105
After cleaning, Line counts = 13
Processing files = 007-3.cha
Line counts = 85
After cleaning, Line counts = 9
Processing files = 010-0.cha
Line counts = 186
After cleaning, Line counts = 22
Processing files = 010-1.cha
Line counts = 201
After cleaning, Line counts = 27
Processing files = 010-2.cha
Line counts = 208
After cleaning, Line counts = 32
Processing files = 010-3.cha
Line counts = 128
After cleaning, Line counts = 17
Processing files = 010-4.cha
Line counts = 92
After cleaning, Line counts = 9
Processing files = 014-2.cha
Line counts = 161
After cleaning, Line counts = 17
Processing files = 016-0.cha
Line counts = 65
After cleaning, Line counts = 6
Processing files = 016-1.cha
Line counts = 57
After cleaning, Line counts = 8
Processing files = 016-3.cha
Line counts = 62
After cleaning, Line counts = 5
Processing files = 016-4.cha
Line counts = 108
After cleaning, Line counts = 13
Processing files = 018-0.cha
Line counts = 199
After cleaning, Line counts = 21
Processing files = 023-0.cha
Line counts = 91
```

```
After cleaning, Line counts = 10
Processing files = 023-2.cha
Line counts = 129
After cleaning, Line counts = 14
Processing files = 024-1.cha
Line counts = 68
After cleaning, Line counts = 8
Processing files = 024-2.cha
Line counts = 91
After cleaning, Line counts = 12
Processing files = 029-0.cha
Line counts = 114
After cleaning, Line counts = 15
Processing files = 029-1.cha
Line counts = 215
After cleaning, Line counts = 27
Processing files = 030-0.cha
Line counts = 148
After cleaning, Line counts = 19
Processing files = 030-1.cha
Line counts = 125
After cleaning, Line counts = 12
Processing files = 033-0.cha
Line counts = 87
After cleaning, Line counts = 7
Processing files = 033-1.cha
Line counts = 124
After cleaning, Line counts = 13
Processing files = 033-2.cha
Line counts = 95
After cleaning, Line counts = 14
Processing files = 033-3.cha
Line counts = 82
After cleaning, Line counts = 9
Processing files = 033-4.cha
Line counts = 118
After cleaning, Line counts = 10
Processing files = 035-0.cha
Line counts = 90
After cleaning, Line counts = 11
Processing files = 035-1.cha
Line counts = 90
After cleaning, Line counts = 13
Processing files = 039-0.cha
Line counts = 81
After cleaning, Line counts = 9
Processing files = 043-0.cha
Line counts = 103
After cleaning, Line counts = 13
Processing files = 046-0.cha
Line counts = 113
After cleaning, Line counts = 16
Processing files = 046-2.cha
Line counts = 134
After cleaning, Line counts = 14
Processing files = 049-0.cha
```

```
Line counts = 153
After cleaning, Line counts = 17
Processing files = 049-1.cha
Line counts = 113
After cleaning, Line counts = 15
Processing files = 050-0.cha
Line counts = 88
After cleaning, Line counts = 12
Processing files = 051-0.cha
Line counts = 103
After cleaning, Line counts = 14
Processing files = 051-1.cha
Line counts = 90
After cleaning, Line counts = 12
Processing files = 051-2.cha
Line counts = 207
After cleaning, Line counts = 27
Processing files = 051-3.cha
Line counts = 151
After cleaning, Line counts = 24
Processing files = 053-1.cha
Line counts = 238
After cleaning, Line counts = 25
Processing files = 057-0.cha
Line counts = 91
After cleaning, Line counts = 12
Processing files = 057-1.cha
Line counts = 96
After cleaning, Line counts = 15
Processing files = 057-2.cha
Line counts = 149
After cleaning, Line counts = 24
Processing files = 058-0.cha
Line counts = 77
After cleaning, Line counts = 8
Processing files = 058-1.cha
Line counts = 119
After cleaning, Line counts = 15
Processing files = 058-3.cha
Line counts = 52
After cleaning, Line counts = 5
Processing files = 058-4.cha
Line counts = 75
After cleaning, Line counts = 5
Processing files = 061-0.cha
Line counts = 165
After cleaning, Line counts = 17
Processing files = 061-1.cha
Line counts = 82
After cleaning, Line counts = 11
Processing files = 062-0.cha
Line counts = 72
After cleaning, Line counts = 8
Processing files = 065-0.cha
Line counts = 100
After cleaning, Line counts = 12
```

```
Processing files = 065-1.cha
Line counts = 115
After cleaning, Line counts = 14
Processing files = 065-2.cha
Line counts = 164
After cleaning, Line counts = 22
Processing files = 066-0.cha
Line counts = 113
After cleaning, Line counts = 12
Processing files = 067-1.cha
Line counts = 103
After cleaning, Line counts = 11
Processing files = 067-2.cha
Line counts = 130
After cleaning, Line counts = 15
Processing files = 070-2.cha
Line counts = 68
After cleaning, Line counts = 9
Processing files = 076-0.cha
Line counts = 172
After cleaning, Line counts = 18
Processing files = 076-2.cha
Line counts = 235
After cleaning, Line counts = 28
Processing files = 076-4.cha
Line counts = 101
After cleaning, Line counts = 11
Processing files = 078-0.cha
Line counts = 127
After cleaning, Line counts = 16
Processing files = 078-1.cha
Line counts = 132
After cleaning, Line counts = 17
Processing files = 087-1.cha
Line counts = 105
After cleaning, Line counts = 12
Processing files = 089-0.cha
Line counts = 136
After cleaning, Line counts = 18
Processing files = 091-0.cha
Line counts = 89
After cleaning, Line counts = 8
Processing files = 091-1.cha
Line counts = 106
After cleaning, Line counts = 9
Processing files = 091-2.cha
Line counts = 73
After cleaning, Line counts = 7
Processing files = 094-1.cha
Line counts = 130
After cleaning, Line counts = 16
Processing files = 094-2.cha
Line counts = 147
After cleaning, Line counts = 14
Processing files = 094-3.cha
Line counts = 102
```

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After cleaning, Line counts = 14
Processing files = 097-1.cha
Line counts = 77
After cleaning, Line counts = 7
Processing files = 120-0.cha
Line counts = 80
After cleaning, Line counts = 9
Processing files = 120-1.cha
Line counts = 122
After cleaning, Line counts = 13
Processing files = 120-2.cha
Line counts = 82
After cleaning, Line counts = 11
Processing files = 120-3.cha
Line counts = 112
After cleaning, Line counts = 12
Processing files = 120-4.cha
Line counts = 88
After cleaning, Line counts = 8
Processing files = 122-0.cha
Line counts = 88
After cleaning, Line counts = 8
Processing files = 122-1.cha
Line counts = 109
After cleaning, Line counts = 10
Processing files = 125-0.cha
Line counts = 116
After cleaning, Line counts = 12
Processing files = 127-0.cha
Line counts = 68
After cleaning, Line counts = 7
Processing files = 134-0.cha
Line counts = 87
After cleaning, Line counts = 10
Processing files = 134-1.cha
Line counts = 55
After cleaning, Line counts = 5
Processing files = 134-2.cha
Line counts = 72
After cleaning, Line counts = 6
Processing files = 134-3.cha
Line counts = 96
After cleaning, Line counts = 9
Processing files = 144-0.cha
Line counts = 129
After cleaning, Line counts = 13
Processing files = 144-1.cha
Line counts = 117
After cleaning, Line counts = 12
Processing files = 148-0.cha
Line counts = 70
After cleaning, Line counts = 9
Processing files = 154-0.cha
Line counts = 109
After cleaning, Line counts = 12
Processing files = 154-1.cha
```

```
Line counts = 84
After cleaning, Line counts = 5
Processing files = 157-0.cha
Line counts = 124
After cleaning, Line counts = 14
Processing files = 157-1.cha
Line counts = 276
After cleaning, Line counts = 33
Processing files = 157-2.cha
Line counts = 92
After cleaning, Line counts = 10
Processing files = 164-1.cha
Line counts = 60
After cleaning, Line counts = 8
Processing files = 164-2.cha
Line counts = 46
After cleaning, Line counts = 6
Processing files = 164-3.cha
Line counts = 68
After cleaning, Line counts = 8
Processing files = 168-0.cha
Line counts = 32
After cleaning, Line counts = 2
Processing files = 168-1.cha
Line counts = 113
After cleaning, Line counts = 14
Processing files = 172-1.cha
Line counts = 85
After cleaning, Line counts = 10
Processing files = 172-2.cha
Line counts = 52
After cleaning, Line counts = 4
Processing files = 172-3.cha
Line counts = 80
After cleaning, Line counts = 11
Processing files = 173-1.cha
Line counts = 120
After cleaning, Line counts = 8
Processing files = 178-0.cha
Line counts = 67
After cleaning, Line counts = 6
Processing files = 178-1.cha
Line counts = 93
After cleaning, Line counts = 12
Processing files = 181-0.cha
Line counts = 73
After cleaning, Line counts = 9
Processing files = 181-1.cha
Line counts = 146
After cleaning, Line counts = 14
Processing files = 181-2.cha
Line counts = 68
After cleaning, Line counts = 7
Processing files = 181-3.cha
Line counts = 131
After cleaning, Line counts = 14
```

```
Processing files = 183-0.cha
Line counts = 60
After cleaning, Line counts = 7
Processing files = 183-1.cha
Line counts = 77
After cleaning, Line counts = 9
Processing files = 183-2.cha
Line counts = 84
After cleaning, Line counts = 7
Processing files = 183-3.cha
Line counts = 78
After cleaning, Line counts = 9
Processing files = 184-0.cha
Line counts = 83
After cleaning, Line counts = 10
Processing files = 184-1.cha
Line counts = 148
After cleaning, Line counts = 16
Processing files = 184-2.cha
Line counts = 97
After cleaning, Line counts = 6
Processing files = 203-0.cha
Line counts = 132
After cleaning, Line counts = 15
Processing files = 203-1.cha
Line counts = 93
After cleaning, Line counts = 10
Processing files = 205-1.cha
Line counts = 170
After cleaning, Line counts = 22
Processing files = 206-0.cha
Line counts = 182
After cleaning, Line counts = 26
Processing files = 207-0.cha
Line counts = 508
After cleaning, Line counts = 65
Processing files = 212-0.cha
Line counts = 134
After cleaning, Line counts = 19
Processing files = 212-1.cha
Line counts = 85
After cleaning, Line counts = 12
Processing files = 212-2.cha
Line counts = 161
After cleaning, Line counts = 21
Processing files = 212-3.cha
Line counts = 129
After cleaning, Line counts = 19
Processing files = 213-1.cha
Line counts = 162
After cleaning, Line counts = 18
Processing files = 213-2.cha
Line counts = 151
After cleaning, Line counts = 19
Processing files = 213-3.cha
Line counts = 136
```

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After cleaning, Line counts = 12
Processing files = 216-0.cha
Line counts = 65
After cleaning, Line counts = 9
Processing files = 216-1.cha
Line counts = 111
After cleaning, Line counts = 12
Processing files = 218-0.cha
Line counts = 159
After cleaning, Line counts = 17
Processing files = 218-1.cha
Line counts = 226
After cleaning, Line counts = 20
Processing files = 220-0.cha
Line counts = 118
After cleaning, Line counts = 16
Processing files = 220-1.cha
Line counts = 88
After cleaning, Line counts = 8
Processing files = 221-0.cha
Line counts = 178
After cleaning, Line counts = 24
Processing files = 221-1.cha
Line counts = 100
After cleaning, Line counts = 10
Processing files = 221-2.cha
Line counts = 104
After cleaning, Line counts = 12
Processing files = 221-3.cha
Line counts = 91
After cleaning, Line counts = 11
Processing files = 222-0.cha
Line counts = 126
After cleaning, Line counts = 10
Processing files = 222-1.cha
Line counts = 142
After cleaning, Line counts = 15
Processing files = 223-0.cha
Line counts = 145
After cleaning, Line counts = 19
Processing files = 223-1.cha
Line counts = 91
After cleaning, Line counts = 10
Processing files = 226-0.cha
Line counts = 120
After cleaning, Line counts = 17
Processing files = 234-0.cha
Line counts = 86
After cleaning, Line counts = 9
Processing files = 235-0.cha
Line counts = 175
After cleaning, Line counts = 19
Processing files = 235-2.cha
Line counts = 104
After cleaning, Line counts = 13
Processing files = 236-0.cha
```

```
Line counts = 126
After cleaning, Line counts = 12
Processing files = 237-1.cha
Line counts = 103
After cleaning, Line counts = 10
Processing files = 237-2.cha
Line counts = 131
After cleaning, Line counts = 18
Processing files = 238-0.cha
Line counts = 131
After cleaning, Line counts = 16
Processing files = 244-0.cha
Line counts = 170
After cleaning, Line counts = 25
Processing files = 247-0.cha
Line counts = 177
After cleaning, Line counts = 16
Processing files = 252-0.cha
Line counts = 68
After cleaning, Line counts = 7
Processing files = 252-1.cha
Line counts = 84
After cleaning, Line counts = 10
Processing files = 252-2.cha
Line counts = 77
After cleaning, Line counts = 4
Processing files = 257-0.cha
Line counts = 162
After cleaning, Line counts = 23
Processing files = 257-2.cha
Line counts = 184
After cleaning, Line counts = 25
Processing files = 260-1.cha
Line counts = 78
After cleaning, Line counts = 8
Processing files = 260-2.cha
Line counts = 85
After cleaning, Line counts = 8
Processing files = 264-0.cha
Line counts = 137
After cleaning, Line counts = 17
Processing files = 268-0.cha
Line counts = 255
After cleaning, Line counts = 27
Processing files = 269-0.cha
Line counts = 149
After cleaning, Line counts = 17
Processing files = 269-1.cha
Line counts = 132
After cleaning, Line counts = 14
Processing files = 270-0.cha
Line counts = 148
After cleaning, Line counts = 18
Processing files = 270-1.cha
Line counts = 84
After cleaning, Line counts = 11
```

```
Processing files = 270-2.cha
Line counts = 108
After cleaning, Line counts = 14
Processing files = 271-2.cha
Line counts = 54
After cleaning, Line counts = 4
Processing files = 276-0.cha
Line counts = 233
After cleaning, Line counts = 30
Processing files = 279-0.cha
Line counts = 171
After cleaning, Line counts = 25
Processing files = 279-1.cha
Line counts = 67
After cleaning, Line counts = 10
Processing files = 282-0.cha
Line counts = 99
After cleaning, Line counts = 12
Processing files = 282-1.cha
Line counts = 88
After cleaning, Line counts = 13
Processing files = 282-2.cha
Line counts = 111
After cleaning, Line counts = 12
Processing files = 283-0.cha
Line counts = 141
After cleaning, Line counts = 21
Processing files = 283-1.cha
Line counts = 99
After cleaning, Line counts = 13
Processing files = 289-2.cha
Line counts = 72
After cleaning, Line counts = 9
Processing files = 291-1.cha
Line counts = 71
After cleaning, Line counts = 7
Processing files = 291-2.cha
Line counts = 74
After cleaning, Line counts = 8
Processing files = 293-1.cha
Line counts = 100
After cleaning, Line counts = 14
Processing files = 306-0.cha
Line counts = 117
After cleaning, Line counts = 17
Processing files = 310-0.cha
Line counts = 98
After cleaning, Line counts = 13
Processing files = 311-0.cha
Line counts = 71
After cleaning, Line counts = 7
Processing files = 319-0.cha
Line counts = 183
After cleaning, Line counts = 18
Processing files = 325-0.cha
Line counts = 92
```

```
After cleaning, Line counts = 8
Processing files = 325-1.cha
Line counts = 77
After cleaning, Line counts = 7
Processing files = 329-0.cha
Line counts = 196
After cleaning, Line counts = 27
Processing files = 334-0.cha
Line counts = 212
After cleaning, Line counts = 32
Processing files = 334-1.cha
Line counts = 152
After cleaning, Line counts = 17
Processing files = 337-0.cha
Line counts = 96
After cleaning, Line counts = 13
Processing files = 338-0.cha
Line counts = 42
After cleaning, Line counts = 6
Processing files = 339-0.cha
Line counts = 79
After cleaning, Line counts = 12
Processing files = 341-0.cha
Line counts = 91
After cleaning, Line counts = 10
Processing files = 342-0.cha
Line counts = 102
After cleaning, Line counts = 16
Processing files = 342-1.cha
Line counts = 107
After cleaning, Line counts = 15
Processing files = 343-0.cha
Line counts = 110
After cleaning, Line counts = 14
Processing files = 344-0.cha
Line counts = 87
After cleaning, Line counts = 9
Processing files = 344-2.cha
Line counts = 149
After cleaning, Line counts = 14
Processing files = 346-0.cha
Line counts = 108
After cleaning, Line counts = 15
Processing files = 349-0.cha
Line counts = 94
After cleaning, Line counts = 12
Processing files = 349-1.cha
Line counts = 116
After cleaning, Line counts = 17
Processing files = 350-0.cha
Line counts = 185
After cleaning, Line counts = 24
Processing files = 350-1.cha
Line counts = 214
After cleaning, Line counts = 27
Processing files = 354-0.cha
```

```
Line counts = 104
After cleaning, Line counts = 14
Processing files = 355-0.cha
Line counts = 94
After cleaning, Line counts = 12
Processing files = 355-1.cha
Line counts = 122
After cleaning, Line counts = 16
Processing files = 356-0.cha
Line counts = 159
After cleaning, Line counts = 13
Processing files = 356-1.cha
Line counts = 97
After cleaning, Line counts = 12
Processing files = 357-0.cha
Line counts = 98
After cleaning, Line counts = 16
Processing files = 358-0.cha
Line counts = 72
After cleaning, Line counts = 10
Processing files = 358-1.cha
Line counts = 112
After cleaning, Line counts = 15
Processing files = 360-0.cha
Line counts = 83
After cleaning, Line counts = 12
Processing files = 361-0.cha
Line counts = 90
After cleaning, Line counts = 11
Processing files = 362-0.cha
Line counts = 126
After cleaning, Line counts = 13
Processing files = 362-1.cha
Line counts = 78
After cleaning, Line counts = 8
Processing files = 368-0.cha
Line counts = 80
After cleaning, Line counts = 11
Processing files = 369-0.cha
Line counts = 93
After cleaning, Line counts = 9
Processing files = 381-0.cha
Line counts = 70
After cleaning, Line counts = 11
Processing files = 381-1.cha
Line counts = 83
After cleaning, Line counts = 9
Processing files = 450-0.cha
Line counts = 99
After cleaning, Line counts = 13
Processing files = 450-1.cha
Line counts = 93
After cleaning, Line counts = 9
Processing files = 458-0.cha
Line counts = 91
After cleaning, Line counts = 12
```

```
Processing files = 461-0.cha
Line counts = 127
After cleaning, Line counts = 19
Processing files = 465-0.cha
Line counts = 61
After cleaning, Line counts = 8
Processing files = 466-0.cha
Line counts = 73
After cleaning, Line counts = 11
Processing files = 466-1.cha
Line counts = 76
After cleaning, Line counts = 6
Processing files = 468-0.cha
Line counts = 76
After cleaning, Line counts = 11
Processing files = 470-1.cha
Line counts = 190
After cleaning, Line counts = 22
Processing files = 471-0.cha
Line counts = 34
After cleaning, Line counts = 5
Processing files = 472-0.cha
Line counts = 74
After cleaning, Line counts = 7
Processing files = 474-0.cha
Line counts = 86
After cleaning, Line counts = 7
Processing files = 476-0.cha
Line counts = 140
After cleaning, Line counts = 20
Processing files = 488-0.cha
Line counts = 87
After cleaning, Line counts = 7
Processing files = 488-1.cha
Line counts = 71
After cleaning, Line counts = 5
Processing files = 492-0.cha
Line counts = 139
After cleaning, Line counts = 13
Processing files = 493-0.cha
Line counts = 150
After cleaning, Line counts = 20
Processing files = 493-1.cha
Line counts = 128
After cleaning, Line counts = 18
Processing files = 497-0.cha
Line counts = 106
After cleaning, Line counts = 14
Processing files = 497-1.cha
Line counts = 122
After cleaning, Line counts = 13
Processing files = 504-0.cha
Line counts = 100
After cleaning, Line counts = 11
Processing files = 506-0.cha
Line counts = 90
```

```
After cleaning, Line counts = 10
Processing files = 508-0.cha
Line counts = 136
After cleaning, Line counts = 17
Processing files = 508-1.cha
Line counts = 122
After cleaning, Line counts = 17
Processing files = 511-0.cha
Line counts = 118
After cleaning, Line counts = 11
Processing files = 515-1.cha
Line counts = 96
After cleaning, Line counts = 10
Processing files = 526-1.cha
Line counts = 139
After cleaning, Line counts = 14
Processing files = 527-0.cha
Line counts = 89
After cleaning, Line counts = 11
Processing files = 527-1.cha
Line counts = 78
After cleaning, Line counts = 7
Processing files = 528-0.cha
Line counts = 96
After cleaning, Line counts = 9
Processing files = 529-0.cha
Line counts = 78
After cleaning, Line counts = 7
Processing files = 530-0.cha
Line counts = 94
After cleaning, Line counts = 11
Processing files = 539-0.cha
Line counts = 185
After cleaning, Line counts = 23
Processing files = 544-0.cha
Line counts = 74
After cleaning, Line counts = 8
Processing files = 544-1.cha
Line counts = 62
After cleaning, Line counts = 6
Processing files = 551-0.cha
Line counts = 159
After cleaning, Line counts = 20
Processing files = 559-0.cha
Line counts = 88
After cleaning, Line counts = 10
Processing files = 562-0.cha
Line counts = 93
After cleaning, Line counts = 12
Processing files = 563-0.cha
Line counts = 127
After cleaning, Line counts = 12
Processing files = 573-0.cha
Line counts = 136
After cleaning, Line counts = 17
Processing files = 578-0.cha
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```
Line counts = 135
After cleaning, Line counts = 17
Processing files = 579-0.cha
Line counts = 61
After cleaning, Line counts = 6
Processing files = 580-0.cha
Line counts = 120
After cleaning, Line counts = 10
Processing files = 581-0.cha
Line counts = 99
After cleaning, Line counts = 8
Processing files = 585-0.cha
Line counts = 114
After cleaning, Line counts = 13
Processing files = 587-0.cha
Line counts = 106
After cleaning, Line counts = 11
Processing files = 591-0.cha
Line counts = 125
After cleaning, Line counts = 13
Processing files = 592-0.cha
Line counts = 107
After cleaning, Line counts = 16
Processing files = 594-0.cha
Line counts = 78
After cleaning, Line counts = 6
Processing files = 595-0.cha
Line counts = 88
After cleaning, Line counts = 13
Processing files = 598-0.cha
Line counts = 61
After cleaning, Line counts = 7
Processing files = 601-0.cha
Line counts = 65
After cleaning, Line counts = 6
Processing files = 607-0.cha
Line counts = 123
After cleaning, Line counts = 15
Processing files = 609-0.cha
Line counts = 127
After cleaning, Line counts = 10
Processing files = 610-0.cha
Line counts = 86
After cleaning, Line counts = 8
Processing files = 615-0.cha
Line counts = 55
After cleaning, Line counts = 6
Processing files = 620-0.cha
Line counts = 128
After cleaning, Line counts = 17
Processing files = 624-0.cha
Line counts = 69
After cleaning, Line counts = 6
Processing files = 635-0.cha
Line counts = 67
After cleaning, Line counts = 5
```

```
Processing files = 636-0.cha
Line counts = 94
After cleaning, Line counts = 7
Processing files = 639-0.cha
Line counts = 78
After cleaning, Line counts = 8
Processing files = 640-0.cha
Line counts = 53
After cleaning, Line counts = 6
Processing files = 642-0.cha
Line counts = 107
After cleaning, Line counts = 14
Processing files = 648-0.cha
Line counts = 88
After cleaning, Line counts = 13
Processing files = 650-0.cha
Line counts = 131
After cleaning, Line counts = 18
Processing files = 651-0.cha
Line counts = 64
After cleaning, Line counts = 5
Processing files = 656-0.cha
Line counts = 48
After cleaning, Line counts = 7
Processing files = 657-0.cha
Line counts = 74
After cleaning, Line counts = 7
Processing files = 660-0.cha
Line counts = 75
After cleaning, Line counts = 8
Processing files = 663-0.cha
Line counts = 55
After cleaning, Line counts = 7
Processing files = 672-0.cha
Line counts = 82
After cleaning, Line counts = 7
Processing files = 674-0.cha
Line counts = 57
After cleaning, Line counts = 5
Processing files = 676-0.cha
Line counts = 72
After cleaning, Line counts = 7
Processing files = 681-0.cha
Line counts = 68
After cleaning, Line counts = 6
Processing files = 689-0.cha
Line counts = 52
After cleaning, Line counts = 6
Processing files = 690-0.cha
Line counts = 84
After cleaning, Line counts = 8
Processing files = 695-0.cha
Line counts = 104
After cleaning, Line counts = 11
Processing files = 698-0.cha
Line counts = 133
```

```
After cleaning, Line counts = 11
Processing files = 702-0.cha
Line counts = 64
After cleaning, Line counts = 7
Processing files = 703-0.cha
Line counts = 86
After cleaning, Line counts = 7
Processing files = 704-0.cha
Line counts = 89
After cleaning, Line counts = 8
Processing files = 705-0.cha
Line counts = 117
After cleaning, Line counts = 12
Processing files = 707-0.cha
Line counts = 106
After cleaning, Line counts = 13
Processing files = 711-0.cha
Line counts = 133
After cleaning, Line counts = 17
Processing files = 714-0.cha
Line counts = 80
After cleaning, Line counts = 4
Processing files = 002-0.cha
Line counts = 125
After cleaning, Line counts = 18
Processing files = 002-1.cha
Line counts = 113
After cleaning, Line counts = 15
Processing files = 002-2.cha
Line counts = 127
After cleaning, Line counts = 16
Processing files = 002-3.cha
Line counts = 132
After cleaning, Line counts = 12
Processing files = 006-2.cha
Line counts = 92
After cleaning, Line counts = 14
Processing files = 006-3.cha
Line counts = 109
After cleaning, Line counts = 10
Processing files = 006-4.cha
Line counts = 134
After cleaning, Line counts = 15
Processing files = 013-0.cha
Line counts = 95
After cleaning, Line counts = 13
Processing files = 013-2.cha
Line counts = 81
After cleaning, Line counts = 13
Processing files = 013-3.cha
Line counts = 92
After cleaning, Line counts = 14
Processing files = 013-4.cha
Line counts = 105
After cleaning, Line counts = 16
Processing files = 015-0.cha
```

```
Line counts = 178
After cleaning, Line counts = 25
Processing files = 015-1.cha
Line counts = 145
After cleaning, Line counts = 21
Processing files = 015-2.cha
Line counts = 214
After cleaning, Line counts = 26
Processing files = 015-3.cha
Line counts = 144
After cleaning, Line counts = 24
Processing files = 015-4.cha
Line counts = 139
After cleaning, Line counts = 19
Processing files = 017-4.cha
Line counts = 206
After cleaning, Line counts = 30
Processing files = 021-0.cha
Line counts = 98
After cleaning, Line counts = 12
Processing files = 021-1.cha
Line counts = 133
After cleaning, Line counts = 12
Processing files = 021-2.cha
Line counts = 71
After cleaning, Line counts = 8
Processing files = 021-3.cha
Line counts = 81
After cleaning, Line counts = 12
Processing files = 021-4.cha
Line counts = 75
After cleaning, Line counts = 8
Processing files = 022-0.cha
Line counts = 103
After cleaning, Line counts = 14
Processing files = 022-1.cha
Line counts = 62
After cleaning, Line counts = 8
Processing files = 022-2.cha
Line counts = 79
After cleaning, Line counts = 8
Processing files = 028-1.cha
Line counts = 147
After cleaning, Line counts = 20
Processing files = 028-4.cha
Line counts = 118
After cleaning, Line counts = 11
Processing files = 034-0.cha
Line counts = 114
After cleaning, Line counts = 12
Processing files = 034-1.cha
Line counts = 64
After cleaning, Line counts = 7
Processing files = 034-2.cha
Line counts = 104
After cleaning, Line counts = 10
```

```
Processing files = 034-3.cha
Line counts = 104
After cleaning, Line counts = 10
Processing files = 034-4.cha
Line counts = 124
After cleaning, Line counts = 13
Processing files = 042-1.cha
Line counts = 137
After cleaning, Line counts = 19
Processing files = 042-2.cha
Line counts = 77
After cleaning, Line counts = 9
Processing files = 042-3.cha
Line counts = 77
After cleaning, Line counts = 11
Processing files = 042-4.cha
Line counts = 68
After cleaning, Line counts = 7
Processing files = 045-0.cha
Line counts = 129
After cleaning, Line counts = 19
Processing files = 045-2.cha
Line counts = 117
After cleaning, Line counts = 19
Processing files = 045-3.cha
Line counts = 110
After cleaning, Line counts = 13
Processing files = 052-0.cha
Line counts = 94
After cleaning, Line counts = 11
Processing files = 052-2.cha
Line counts = 36
After cleaning, Line counts = 4
Processing files = 054-0.cha
Line counts = 104
After cleaning, Line counts = 14
Processing files = 055-0.cha
Line counts = 108
After cleaning, Line counts = 16
Processing files = 056-0.cha
Line counts = 96
After cleaning, Line counts = 9
Processing files = 056-3.cha
Line counts = 66
After cleaning, Line counts = 7
Processing files = 056-4.cha
Line counts = 71
After cleaning, Line counts = 8
Processing files = 059-2.cha
Line counts = 84
After cleaning, Line counts = 10
Processing files = 059-3.cha
Line counts = 70
After cleaning, Line counts = 9
Processing files = 059-4.cha
Line counts = 72
```

```
After cleaning, Line counts = 9
Processing files = 068-0.cha
Line counts = 88
After cleaning, Line counts = 10
Processing files = 068-2.cha
Line counts = 76
After cleaning, Line counts = 11
Processing files = 068-3.cha
Line counts = 98
After cleaning, Line counts = 9
Processing files = 071-0.cha
Line counts = 112
After cleaning, Line counts = 16
Processing files = 071-1.cha
Line counts = 109
After cleaning, Line counts = 18
Processing files = 071-2.cha
Line counts = 123
After cleaning, Line counts = 22
Processing files = 071-3.cha
Line counts = 136
After cleaning, Line counts = 19
Processing files = 071-4.cha
Line counts = 91
After cleaning, Line counts = 13
Processing files = 073-0.cha
Line counts = 111
After cleaning, Line counts = 15
Processing files = 073-1.cha
Line counts = 65
After cleaning, Line counts = 6
Processing files = 073-3.cha
Line counts = 102
After cleaning, Line counts = 10
Processing files = 086-0.cha
Line counts = 226
After cleaning, Line counts = 28
Processing files = 086-1.cha
Line counts = 113
After cleaning, Line counts = 16
Processing files = 086-2.cha
Line counts = 108
After cleaning, Line counts = 12
Processing files = 086-3.cha
Line counts = 128
After cleaning, Line counts = 15
Processing files = 086-4.cha
Line counts = 143
After cleaning, Line counts = 19
Processing files = 092-0.cha
Line counts = 66
After cleaning, Line counts = 7
Processing files = 092-1.cha
Line counts = 102
After cleaning, Line counts = 14
Processing files = 092-2.cha
```

```
Line counts = 51
After cleaning, Line counts = 5
Processing files = 092-3.cha
Line counts = 63
After cleaning, Line counts = 9
Processing files = 093-0.cha
Line counts = 123
After cleaning, Line counts = 20
Processing files = 093-1.cha
Line counts = 75
After cleaning, Line counts = 11
Processing files = 096-1.cha
Line counts = 131
After cleaning, Line counts = 17
Processing files = 096-2.cha
Line counts = 112
After cleaning, Line counts = 18
Processing files = 105-0.cha
Line counts = 114
After cleaning, Line counts = 13
Processing files = 105-1.cha
Line counts = 97
After cleaning, Line counts = 11
Processing files = 105-2.cha
Line counts = 148
After cleaning, Line counts = 14
Processing files = 107-1.cha
Line counts = 85
After cleaning, Line counts = 11
Processing files = 107-2.cha
Line counts = 57
After cleaning, Line counts = 8
Processing files = 109-1.cha
Line counts = 122
After cleaning, Line counts = 19
Processing files = 109-3.cha
Line counts = 103
After cleaning, Line counts = 12
Processing files = 109-4.cha
Line counts = 108
After cleaning, Line counts = 18
Processing files = 113-0.cha
Line counts = 80
After cleaning, Line counts = 12
Processing files = 113-1.cha
Line counts = 87
After cleaning, Line counts = 13
Processing files = 113-2.cha
Line counts = 115
After cleaning, Line counts = 17
Processing files = 113-3.cha
Line counts = 73
After cleaning, Line counts = 8
Processing files = 114-0.cha
Line counts = 100
After cleaning, Line counts = 12
```

```
Processing files = 114-1.cha
Line counts = 74
After cleaning, Line counts = 7
Processing files = 114-2.cha
Line counts = 78
After cleaning, Line counts = 11
Processing files = 114-3.cha
Line counts = 64
After cleaning, Line counts = 7
Processing files = 114-4.cha
Line counts = 83
After cleaning, Line counts = 10
Processing files = 118-0.cha
Line counts = 65
After cleaning, Line counts = 7
Processing files = 118-1.cha
Line counts = 91
After cleaning, Line counts = 10
Processing files = 118-2.cha
Line counts = 69
After cleaning, Line counts = 10
Processing files = 118-3.cha
Line counts = 91
After cleaning, Line counts = 7
Processing files = 118-4.cha
Line counts = 90
After cleaning, Line counts = 8
Processing files = 121-0.cha
Line counts = 193
After cleaning, Line counts = 25
Processing files = 121-1.cha
Line counts = 149
After cleaning, Line counts = 24
Processing files = 121-2.cha
Line counts = 163
After cleaning, Line counts = 21
Processing files = 121-3.cha
Line counts = 143
After cleaning, Line counts = 19
Processing files = 121-4.cha
Line counts = 161
After cleaning, Line counts = 21
Processing files = 124-0.cha
Line counts = 101
After cleaning, Line counts = 15
Processing files = 124-1.cha
Line counts = 128
After cleaning, Line counts = 23
Processing files = 128-1.cha
Line counts = 169
After cleaning, Line counts = 23
Processing files = 128-2.cha
Line counts = 255
After cleaning, Line counts = 34
Processing files = 128-3.cha
Line counts = 331
```

```
After cleaning, Line counts = 41
Processing files = 129-1.cha
Line counts = 53
After cleaning, Line counts = 4
Processing files = 130-1.cha
Line counts = 48
After cleaning, Line counts = 5
Processing files = 130-2.cha
Line counts = 58
After cleaning, Line counts = 9
Processing files = 130-3.cha
Line counts = 65
After cleaning, Line counts = 4
Processing files = 132-0.cha
Line counts = 97
After cleaning, Line counts = 14
Processing files = 132-1.cha
Line counts = 86
After cleaning, Line counts = 11
Processing files = 137-0.cha
Line counts = 149
After cleaning, Line counts = 22
Processing files = 137-1.cha
Line counts = 91
After cleaning, Line counts = 10
Processing files = 137-2.cha
Line counts = 75
After cleaning, Line counts = 12
Processing files = 137-3.cha
Line counts = 112
After cleaning, Line counts = 10
Processing files = 138-1.cha
Line counts = 112
After cleaning, Line counts = 17
Processing files = 138-3.cha
Line counts = 105
After cleaning, Line counts = 10
Processing files = 139-0.cha
Line counts = 111
After cleaning, Line counts = 13
Processing files = 139-1.cha
Line counts = 61
After cleaning, Line counts = 7
Processing files = 139-3.cha
Line counts = 46
After cleaning, Line counts = 5
Processing files = 140-0.cha
Line counts = 148
After cleaning, Line counts = 19
Processing files = 140-3.cha
Line counts = 74
After cleaning, Line counts = 7
Processing files = 141-0.cha
Line counts = 79
After cleaning, Line counts = 11
Processing files = 141-1.cha
```

```
Line counts = 59
After cleaning, Line counts = 7
Processing files = 141-2.cha
Line counts = 84
After cleaning, Line counts = 11
Processing files = 141-3.cha
Line counts = 124
After cleaning, Line counts = 18
Processing files = 142-0.cha
Line counts = 99
After cleaning, Line counts = 12
Processing files = 142-1.cha
Line counts = 80
After cleaning, Line counts = 9
Processing files = 142-3.cha
Line counts = 81
After cleaning, Line counts = 9
Processing files = 143-3.cha
Line counts = 68
After cleaning, Line counts = 9
Processing files = 145-1.cha
Line counts = 87
After cleaning, Line counts = 15
Processing files = 145-3.cha
Line counts = 149
After cleaning, Line counts = 15
Processing files = 146-1.cha
Line counts = 117
After cleaning, Line counts = 14
Processing files = 150-0.cha
Line counts = 95
After cleaning, Line counts = 13
Processing files = 150-1.cha
Line counts = 97
After cleaning, Line counts = 12
Processing files = 150-2.cha
Line counts = 135
After cleaning, Line counts = 16
Processing files = 155-0.cha
Line counts = 65
After cleaning, Line counts = 9
Processing files = 155-2.cha
Line counts = 122
After cleaning, Line counts = 17
Processing files = 155-3.cha
Line counts = 67
After cleaning, Line counts = 8
Processing files = 158-0.cha
Line counts = 129
After cleaning, Line counts = 18
Processing files = 158-1.cha
Line counts = 113
After cleaning, Line counts = 20
Processing files = 158-2.cha
Line counts = 82
After cleaning, Line counts = 16
```

```
Processing files = 158-3.cha
Line counts = 81
After cleaning, Line counts = 10
Processing files = 166-0.cha
Line counts = 56
After cleaning, Line counts = 5
Processing files = 166-1.cha
Line counts = 127
After cleaning, Line counts = 18
Processing files = 166-2.cha
Line counts = 92
After cleaning, Line counts = 13
Processing files = 167-1.cha
Line counts = 83
After cleaning, Line counts = 9
Processing files = 167-2.cha
Line counts = 79
After cleaning, Line counts = 11
Processing files = 167-3.cha
Line counts = 161
After cleaning, Line counts = 21
Processing files = 171-0.cha
Line counts = 61
After cleaning, Line counts = 9
Processing files = 171-1.cha
Line counts = 56
After cleaning, Line counts = 7
Processing files = 172-0.cha
Line counts = 123
After cleaning, Line counts = 17
Processing files = 175-0.cha
Line counts = 142
After cleaning, Line counts = 23
Processing files = 175-1.cha
Line counts = 115
After cleaning, Line counts = 15
Processing files = 175-2.cha
Line counts = 135
After cleaning, Line counts = 17
Processing files = 175-3.cha
Line counts = 73
After cleaning, Line counts = 9
Processing files = 182-3.cha
Line counts = 85
After cleaning, Line counts = 11
Processing files = 192-0.cha
Line counts = 59
After cleaning, Line counts = 8
Processing files = 192-2.cha
Line counts = 121
After cleaning, Line counts = 14
Processing files = 196-0.cha
Line counts = 102
After cleaning, Line counts = 12
Processing files = 196-1.cha
Line counts = 99
```

```
After cleaning, Line counts = 12
Processing files = 208-0.cha
Line counts = 76
After cleaning, Line counts = 11
Processing files = 208-1.cha
Line counts = 63
After cleaning, Line counts = 6
Processing files = 208-2.cha
Line counts = 97
After cleaning, Line counts = 15
Processing files = 209-1.cha
Line counts = 81
After cleaning, Line counts = 9
Processing files = 209-2.cha
Line counts = 109
After cleaning, Line counts = 11
Processing files = 209-3.cha
Line counts = 155
After cleaning, Line counts = 23
Processing files = 210-1.cha
Line counts = 149
After cleaning, Line counts = 22
Processing files = 210-2.cha
Line counts = 159
After cleaning, Line counts = 21
Processing files = 211-1.cha
Line counts = 87
After cleaning, Line counts = 12
Processing files = 211-2.cha
Line counts = 86
After cleaning, Line counts = 12
Processing files = 225-0.cha
Line counts = 134
After cleaning, Line counts = 14
Processing files = 225-2.cha
Line counts = 123
After cleaning, Line counts = 15
Processing files = 227-0.cha
Line counts = 123
After cleaning, Line counts = 19
Processing files = 227-1.cha
Line counts = 146
After cleaning, Line counts = 17
Processing files = 229-1.cha
Line counts = 60
After cleaning, Line counts = 9
Processing files = 229-2.cha
Line counts = 56
After cleaning, Line counts = 8
Processing files = 232-0.cha
Line counts = 127
After cleaning, Line counts = 17
Processing files = 232-1.cha
Line counts = 140
After cleaning, Line counts = 18
Processing files = 242-0.cha
```

```
Line counts = 93
After cleaning, Line counts = 9
Processing files = 242-1.cha
Line counts = 141
After cleaning, Line counts = 21
Processing files = 242-2.cha
Line counts = 167
After cleaning, Line counts = 26
Processing files = 243-0.cha
Line counts = 184
After cleaning, Line counts = 23
Processing files = 243-1.cha
Line counts = 122
After cleaning, Line counts = 16
Processing files = 245-0.cha
Line counts = 82
After cleaning, Line counts = 7
Processing files = 245-1.cha
Line counts = 118
After cleaning, Line counts = 15
Processing files = 245-2.cha
Line counts = 122
After cleaning, Line counts = 13
Processing files = 248-0.cha
Line counts = 125
After cleaning, Line counts = 17
Processing files = 248-1.cha
Line counts = 111
After cleaning, Line counts = 12
Processing files = 248-2.cha
Line counts = 64
After cleaning, Line counts = 7
Processing files = 255-0.cha
Line counts = 84
After cleaning, Line counts = 10
Processing files = 255-1.cha
Line counts = 48
After cleaning, Line counts = 5
Processing files = 256-0.cha
Line counts = 101
After cleaning, Line counts = 12
Processing files = 256-1.cha
Line counts = 66
After cleaning, Line counts = 9
Processing files = 256-2.cha
Line counts = 145
After cleaning, Line counts = 22
Processing files = 266-0.cha
Line counts = 74
After cleaning, Line counts = 9
Processing files = 266-1.cha
Line counts = 74
After cleaning, Line counts = 11
Processing files = 266-2.cha
Line counts = 144
After cleaning, Line counts = 19
```

```
Processing files = 267-0.cha
Line counts = 83
After cleaning, Line counts = 7
Processing files = 267-2.cha
Line counts = 93
After cleaning, Line counts = 11
Processing files = 274-0.cha
Line counts = 107
After cleaning, Line counts = 14
Processing files = 274-1.cha
Line counts = 69
After cleaning, Line counts = 8
Processing files = 274-2.cha
Line counts = 70
After cleaning, Line counts = 5
Processing files = 275-0.cha
Line counts = 127
After cleaning, Line counts = 14
Processing files = 275-1.cha
Line counts = 86
After cleaning, Line counts = 8
Processing files = 280-0.cha
Line counts = 129
After cleaning, Line counts = 19
Processing files = 280-1.cha
Line counts = 53
After cleaning, Line counts = 8
Processing files = 280-2.cha
Line counts = 94
After cleaning, Line counts = 11
Processing files = 292-1.cha
Line counts = 85
After cleaning, Line counts = 13
Processing files = 295-0.cha
Line counts = 68
After cleaning, Line counts = 11
Processing files = 295-1.cha
Line counts = 48
After cleaning, Line counts = 5
Processing files = 296-0.cha
Line counts = 74
After cleaning, Line counts = 6
Processing files = 296-1.cha
Line counts = 131
After cleaning, Line counts = 15
Processing files = 296-2.cha
Line counts = 117
After cleaning, Line counts = 17
Processing files = 297-1.cha
Line counts = 64
After cleaning, Line counts = 9
Processing files = 297-2.cha
Line counts = 88
After cleaning, Line counts = 10
Processing files = 298-1.cha
Line counts = 102
```

```
After cleaning, Line counts = 10
Processing files = 299-1.cha
Line counts = 96
After cleaning, Line counts = 9
Processing files = 302-0.cha
Line counts = 61
After cleaning, Line counts = 9
Processing files = 304-1.cha
Line counts = 45
After cleaning, Line counts = 5
Processing files = 304-2.cha
Line counts = 91
After cleaning, Line counts = 9
Processing files = 318-0.cha
Line counts = 94
After cleaning, Line counts = 10
Processing files = 318-1.cha
Line counts = 78
After cleaning, Line counts = 7
Processing files = 318-2.cha
Line counts = 72
After cleaning, Line counts = 7
Processing files = 322-1.cha
Line counts = 85
After cleaning, Line counts = 9
Processing files = 322-2.cha
Line counts = 67
After cleaning, Line counts = 8
Processing files = 323-0.cha
Line counts = 129
After cleaning, Line counts = 23
Processing files = 323-1.cha
Line counts = 115
After cleaning, Line counts = 12
Processing files = 332-0.cha
Line counts = 48
After cleaning, Line counts = 6
Processing files = 336-1.cha
Line counts = 120
After cleaning, Line counts = 11
Processing files = 340-0.cha
Line counts = 100
After cleaning, Line counts = 15
Processing files = 612-0.cha
Line counts = 107
After cleaning, Line counts = 11
Processing files = 627-0.cha
Line counts = 92
After cleaning, Line counts = 12
Processing files = 631-0.cha
Line counts = 58
After cleaning, Line counts = 8
Processing files = 661-0.cha
Line counts = 98
After cleaning, Line counts = 12
Processing files = 668-0.cha
```

```
Line counts = 112
After cleaning, Line counts = 14
Processing files = 678-0.cha
Line counts = 101
After cleaning, Line counts = 10
Processing files = 684-0.cha
Line counts = 140
After cleaning, Line counts = 20
Processing files = 686-0.cha
Line counts = 124
After cleaning, Line counts = 12
Processing files = 688-0.cha
Line counts = 91
After cleaning, Line counts = 11
Processing files = 691-0.cha
Line counts = 141
After cleaning, Line counts = 19
Processing files = 709-0.cha
Line counts = 88
After cleaning, Line counts = 10
Processing files = 709-2.cha
Line counts = 109
After cleaning, Line counts = 15
```

In [8]:
```python
# double check data
assert len(train_data["ad"]) == 306
assert len(train_data["cn"]) == 243
assert train_data["ad"][0].startswith("mhm")
assert train_data["ad"][1].startswith("mhm")
assert train_data["cn"][0].startswith("the scene is <in the>")
assert train_data["cn"][1].startswith("oh I see the sink is running over")
print("Passed")
```

```
Passed
```

In [9]:
```python
# assert filenames
assert file_names["ad"][0].startswith("001-0")
assert file_names["ad"][1].startswith("001-2")
assert file_names["ad"][-1].startswith("714-0")

assert file_names["cn"][0].startswith("002-0")
assert file_names["cn"][1].startswith("002-1")
assert file_names["cn"][-1].startswith("709-2")

print("Passed")
```

```
Passed
```

In [10]:
```python
# take a look at the data
print(train_data["ad"][0][:100])
```

```
mhm .
alright .
there's &-um a young boy that's getting a cookie jar .
and it  he's &-uh in bad shap
```

In [11]:
```
# take a look at the data
print(train_data["cn"][0][:100])
```

```
the scene is <in the>  in the kitchen .
the mother is wiping dishes and the water is running on the
```

In [12]:
```
print(file_names["ad"][0:5])
```

```
['001-0', '001-2', '003-0', '005-0', '005-2']
```

In [13]:
```
print(file_names["cn"][0:5])
```

```
['002-0', '002-1', '002-2', '002-3', '006-2']
```

In [14]:
```
# build word-clouds
def make_wordcloud(text_list):
    """
    display a wordcloud based on the text list
    parameters:
        text_list: list of text (paragraphs)
    returns:
        list of frequent words
    """
    text = "\n".join(text_list).lower()
    wordcloud = WordCloud().generate(text)

    # process text and return counts of frequent words
    result = sorted(WordCloud().process_text(text).items(),
                    key=lambda item: item[1],
                    reverse=True
                   )


    plt.imshow(wordcloud)
    plt.axis("off")
    plt.show()

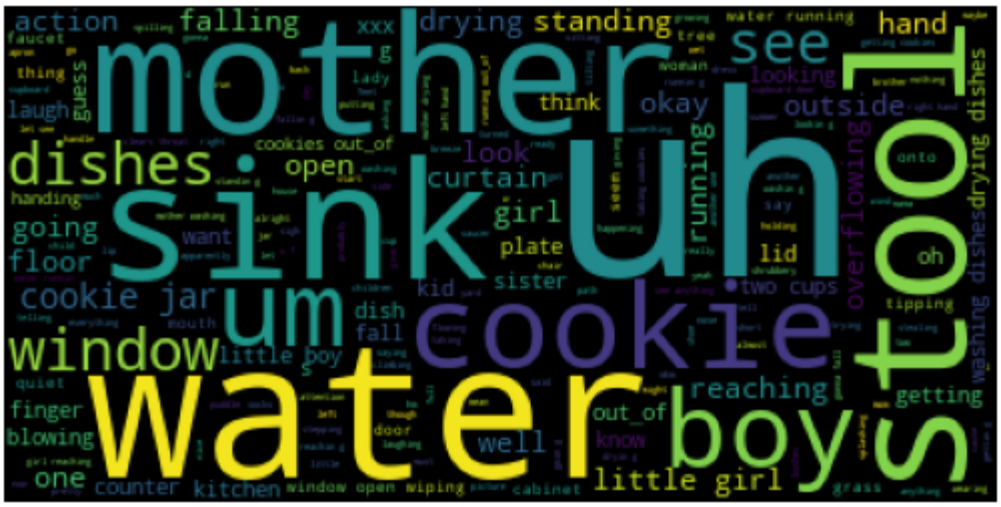
    return result
```

In [15]:
```
# see the most frequent words of AD (positive) group
ad_top_words = make_wordcloud(train_data["ad"])
print(ad_top_words[:50])
ad_top_words_set = set([w
                        for w, count in ad_top_words[:200]
                       ])
```




```
[('uh', 714), ('water', 274), ('cookie', 257), ('sink', 232), ('cookie jar', 217),
('stool', 211), ('mother', 202), ('well', 193), ('dishe', 192), ('boy', 186), ('oh',
159), ('see', 158), ('laugh', 140), ('floor', 123), ('girl', 121), ('little girl', 1
08), ('window', 103), ('look', 103), ('little boy', 103), ('one', 101), ('know', 9
9), ('um', 96), ('xxx', 91), ('kid', 79), ('s', 78), ('hand', 75), ('going', 74),
('got', 74), ('g', 73), ('dryin g', 70), ('okay', 69), ('guess', 68), ('reaching', 6
6), ('g dishes', 65), ('sister', 63), ('think', 62), ('standing', 61), ('falling', 6
0), ('yeah', 58), ('gonna', 57), ('dish', 55), ('getting', 54), ('hm', 53), ('runnin
g', 53), ('little', 52), ('washin g', 52), ('gettin g', 52), ('want', 50), ('thing',
49), ('plate', 49)]
```

In [16]:
```
# see the most frequent words of CN (negative) group
cn_top_words = make_wordcloud(train_data["cn"])
print(cn_top_words[:50])
cn_top_words_set = set([w
                        for w, count in cn_top_words[:200]
                       ])
```



```
[('uh', 527), ('water', 247), ('mother', 243), ('sink', 242), ('stool', 234), ('cook
ie', 227), ('boy', 184), ('um', 184), ('window', 167), ('dishes', 164), ('see', 16
3), ('cookie jar', 162), ('okay', 116), ('standing', 113), ('action', 110), ('look',
108), ('little girl', 107), ('reaching', 104), ('open', 101), ('floor', 98), ('gir
l', 96), ('one', 92), ('curtain', 91), ('falling', 89), ('hand', 83), ('well', 80),
('overflowing', 79), ('drying', 78), ('outside', 77), ('going', 72), ('running', 7
1), ('little boy', 67), ('plate', 65), ('g', 62), ('oh', 57), ('dish', 56), ('siste
r', 56), ('getting', 55), ('fall', 54), ('know', 54), ('drying dishes', 53), ('kitch
en', 49), ('want', 49), ('guess', 49), ('s', 44), ('washing dishes', 44), ('lookin
g', 43), ('out_of', 43), ('two cups', 43), ('counter', 42)]
```

In [17]:
```python
# observe the difference in 2 word set
ad_unique = ad_top_words_set - cn_top_words_set
cn_unique = cn_top_words_set - ad_top_words_set
print("AD unique words =\n", ad_unique)
print("----------------")
print("CN unique words =\n", cn_unique)
```

```
AD unique words =
 {'fell', 'wa', 'tryin g', 'somethin g', 'upset', 'even', 'taking cookie', 'f', 'g d
ishes', 'comin g', 'huh', 'inv okay', 'whatever', 'you_know', 'spilled', 'part', 'i_
mean', 'young', 'water spilling', 'spigot', 'way', 'good', 'dishe', 'th', 'yet',
'c', 'foot', 'hafta', 'doin g', 'th em', 'coming', 'somebody', 'spillin g', 'still',
'three', 'u', 'out_o f', 'hurt', 'yes', 'turn', 'water run', 'sh', 'wash', 'call',
'time', 'laughin g', 'two cup', 'take', 't', 'ladder', 'already'}
----------------
CN unique words =
 {'blowing', 'shrubbery', 'laughing', 'lip', 'growing', 'taking cookies', 'bushes',
'giving', 'taking', 'girl reaching', 'nose', 'really', 'saying', 'mother washing',
'pretty', 'getting cookies', 'stepping', 'o f', 'wind', 'socks', 'start', 'almost',
'summer', 'attention', 'splashing', 'two cups', 'handle', 'running out_of', 'turne
d', 'window open', 'short', 'full', 'asking', 'day', 'nothing', 'breeze', 'tilting',
'mean', 'left hand', 'sitting', 'sigh', 'climbing', 'dress', 'child', 'dishes', 'wea
ring', 'right hand', 'another one', 'mother drying', 'putting', 'puddle'}
```

In [18]:
```python
# combine data into a dataframe, for export to a csv for further processing
train_dict = {
    "file_id": [],  #  file name
    "text": [],    # the transcript
    "target": []   # the target: positive (AD), negative (CN)
}
for item, fname in zip(train_data["ad"], file_names["ad"]):
    train_dict["file_id"].append(fname)
    train_dict["text"].append(item)
    train_dict["target"].append("AD")

for item, fname in zip(train_data["cn"], file_names["cn"]):
    train_dict["file_id"].append(fname)
    train_dict["text"].append(item)
    train_dict["target"].append("CN")

print("Text count =", len(train_dict["text"]))
print("Target count =", len(train_dict["target"]))
print("File_id count =", len(train_dict["target"]))
```

```
Text count = 549
Target count = 549
File_id count = 549
```

In [19]:
```
train_df = pd.DataFrame(train_dict)
print(len(train_df))
train_df.head()
```

```
549
```

Out[19]:

	file_id	text	target
0	001-0	mhm .\nalright .\nthere's &-um a young boy tha...	AD
1	001-2	mhm .\nthere's a young boy &-uh going in a coo...	AD
2	003-0	here's a cookie jar .\nand the lid is off the ...	AD
3	005-0	the boy is slipping off the stool .\nhe's tryi...	AD
4	005-2	okay ‡ he's fallin(g) off a chair .\nshe's &-...	AD

In [20]:
```
train_df.describe()
```

Out[20]:

	file_id	text	target
count	549	549	549
unique	549	549	2
top	001-0	mhm .\nalright .\nthere's &-um a young boy tha...	AD
freq	1	1	306

We can see we have total of 549 examples, with 306 (about 55%) positive (AD) and 243 (about 45%) negative (CN). The train_set is relatively balanced, although there are slightly more examples with positive labels.

Create CSV output

In [22]:
```
# write to csv
print("Writing to csv files...")
path = "../data/splits/"
train_df.to_csv(path + "train_pitt.csv", index=False)
print("Completed")
```

```
Writing to csv files...
Completed
```